

Quasi-Eigentone Framework v0.1 (#397) — unifying SM unstable particles as quasi-eigentones; decay = Green coproduct overlap with lower states; the structural distinction between stable and unstable particles is excited-vs-ground in K-type radial towers. Generalizes Elie's E3 β -decay to the full SM.

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2026-05-30 Saturday 15:30 EDT

Quasi-Eigentone Framework v0.1 — unifying SM unstable particles

0. The question

Why are some particles stable (electron, proton, photon, neutrinos) while others decay (muon, tau, neutron, W/Z, Higgs, hadrons)? The Standard Model treats this as a per-particle empirical fact — each unstable particle's lifetime is a derived consequence of specific Feynman diagrams and phase space. The substrate framework asks: is there a single STRUCTURAL distinction that places every SM particle in a stable-or-unstable class, derivable from the engine?

This framework gives that distinction: stable = true eigentone (ground state in its K-type sector); unstable = quasi-eigentone (excited state, decays via Green coproduct overlap with lower states).

1. Definition: true eigentone vs quasi-eigentone

The substrate's holomorphic discrete series organizes physical states into K-type towers. Within each K-type sector (e.g., the lepton sector $V_{(1/2,1/2)}$ or the bulk-composite sector at $V_{(1,1)_{\text{bulk}}}$), states form a radial tower indexed by mass/Casimir level.

TRUE EIGENTONE (stable): the GROUND STATE in its K-type sector. Either: - The lowest-mass state in its sector (e.g., electron is the lightest charged lepton in the $V_{(1/2,1/2)}$ lepton sector). - Massless (e.g., photon at $V_{(1,0)}$) — always at the bottom of the Casimir tower. - A confined eigentone-in-vacuum (e.g., gluon — stable as a substrate mode, but never propagates in isolation due to bulk-color confinement).

True eigentones have no allowed decay channels — there is no lower-energy state in their K-type sector + grading-allowed coproduct decomposition.

QUASI-EIGENTONE (unstable): an EXCITED state in its K-type sector. Has: - A nonzero overlap with the coproduct decomposition into lower-energy states. - Grading-conserved decay channels (the Green coproduct preserves the Cartan grading = the Q, B, L charges). - A finite lifetime determined by the overlap $\times$ phase space $\times$ kernel matrix elements.

Quasi-eigentones decay because the engine's Green coproduct provides a structural mechanism for their decomposition into kinematically-allowed lower states.

2. The decay mechanism (engine-level)

From engine v0.1 (Lyra) + engine v0.2 (Elie), the Green coproduct on the substrate Hall algebra is:

$$\Delta(u_X) = \sum q^{\langle \dim M, \dim L \rangle} (a_M a_L / a_X) g^X_{\{ML\}} u_M \otimes u_L$$

where u_X is a Hall-algebra element (a K-type's algebraic image) and $\Delta(u_X)$ decomposes it into pairs (u_M, u_L) of lower-degree elements. Physically:

- $\Delta(\text{quasi-eigentone}) = \Sigma (\text{allowed decay channels})$ — each nonzero coproduct component is a kinematically-allowed decay channel.
- Conservation = grading on the Hopf algebra — the Cartan grading preserves Q, B, L (electric charge, baryon number, lepton number) automatically.
- Decay rate $\propto |\text{overlap}|^2 \times \text{phase-space} \times \text{kernel-matrix-element}$ — the overlap is the coproduct coefficient (structural); phase-space and kernel matrix-elements give the mass-dependence (L4 open).

β -decay (Elie E3) is the first example, generalized: the neutron K-type's Green coproduct gives $n \rightarrow p + e + \bar{\nu}$ as the leading decomposition with Q, B, L conservation. This framework GENERALIZES: every SM decay is the Green-coproduct decomposition of the quasi-eigentone's K-type into kinematically-allowed lower K-types.

3. Classification of SM particles

Particle	K-type / sector	Mass (MeV)	Class	Structural reason
electron	$V_{(1/2,1/2)}$ gen-1	0.511	TRUE EIGENTONE	lightest charged lepton in $V_{(1/2,1/2)}$ sector
muon	$V_{(1/2,1/2)}$ gen-2	105.7	QUASI	gen-2 above gen-1; coproduct decay $\mu \rightarrow e + \bar{\nu}_e + \nu_\mu$
tau	$V_{(1/2,1/2)}$ gen-3	1777	QUASI	gen-3 above gens 1-2; many decay channels (hadronic + leptonic)
neutrinos	$V_{(1/2,1/2)}$ neutral	$\sim 10^{-3}$ eV	TRUE EIGENTONE	lightest in neutrino sub-sector (Dirac, per F5)
photon	$V_{(1,0)}$	0	TRUE EIGENTONE	massless gauge; always at bottom of Casimir tower
gluon	$V_{(1,1)}$ color	0	EIGENTONE-IN-VACUUM	massless gauge; bulk-color confinement prevents isolated propagation
W	$V_{(1,1)}$ charged	80,400	QUASI	massive gauge (EWSB); coproduct decay $W \rightarrow \ell + \nu$ or $q + \bar{q}'$
Z	$V_{(1,1)}$ neutral	91,200	QUASI	massive gauge (EWSB); coproduct decay $Z \rightarrow f + \bar{f}$
Higgs	$V_{(0,0)}$	125,000	QUASI	massive scalar above vacuum; decays to fermion pairs / gluon pairs

Particle	K-type / sector	Mass (MeV)	Class	Structural reason
proton	V_(1,1) bulk k=6	938	TRUE EIGENTONE	lightest baryon at C_2 = 6 closure
neutron	V_(1,1) bulk k=6 isospin partner	940	QUASI	1.3 MeV above proton; coproduct decay $n \rightarrow p + e + \bar{\nu}$ (E3)
pion	meson	140	QUASI	meson; coproduct decay $\pi^+ \rightarrow \mu + \nu$
K-mesons	V-compound meson V-compound	~500	QUASI	strange meson; multiple decay channels
Λ baryon	bulk V-compound	1116	QUASI	strange baryon; decays to nucleon + π
(other hadrons)	various bulk K-types	various	QUASI	excited bulk states; decay to lower bulk states

The structural pattern: - TRUE EIGENTONES: electron, neutrinos, photon, proton — the bottoms of their respective K-type sectors. - EIGENTONES-IN-VACUUM: gluon (massless but confined). - QUASI-EIGENTONES: everything else — excited radial/generation/isospin levels with structural coproduct decay channels.

4. Why some particles are stable: the structural reason

The substrate's K-type structure organizes states into towers. The bottom of each tower is a true eigentone (no allowed decay). The K-type “stable particles” of the SM correspond exactly to the bottoms of their K-type sectors:

- Electron = lowest in V_(1/2,1/2) lepton charged sector.
- Electron neutrino = lowest in V_(1/2,1/2) lepton neutral sector (Dirac, per F5 prediction).
- Photon = V_(1,0) vector at zero Casimir restriction (massless).
- Proton = lowest baryon in V_(1,1) bulk (C_2 = 6 closure; lightest 3-quark color-singlet).
- Gluon = massless but confined — eigentone-in-vacuum.

The SM's stable-particle set is NOT an empirical accident — it's the bottoms-of-K-type-sectors prediction of the substrate. Any new “stable” particle would have to be a new bottom in some K-type sector; the Periodic Table predicts none beyond the above.

5. Connection to L4 mass framework

L4 v0.1 placed the lepton at Casimir $5/2 = \rho_1$ (the natural mass-setting unit). For quasi-eigentones, the mass spacing within a K-type sector determines the decay phase space:

- Lepton sector spacing: $m_\mu/m_e = (24/\pi^2)^6$ (T190); $m_\tau/m_\mu \approx 17$ (T2003). The radial-tower spacing within the V_(1/2,1/2) sector.
- Bulk-composite spacing: m_p, m_n , hadron spectroscopy — the radial tower in V_(1,1)_bulk.
- Gauge sector spacing: m_W, m_Z , Higgs — via EWSB and Higgs VEV ($\sin^2\theta_W = 2/9$, $m_W/m_Z = \sqrt{(g/N_c^2)}$).

Decay rates depend on the specific MASS RATIOS within and across K-type sectors, which is L4 v0.2 multi-week work. This framework places the STRUCTURAL distinction (eigentone vs quasi-eigentone); the QUANTITATIVE decay rates require the bulk K-type radial tower computation (Elie's lane after affine pin).

6. Connection to the Movie program (V0/V3)

The Movie program (Casey + team) visualizes substrate operations: - V0 spectral score: shows the K-type Casimir spectrum with each particle at its K-type level. True eigentones at sector bottoms; quasi-eigentones at radial-tower upper levels. - V3 formation/decay movies: show particle formation (build up from coproduct multiplication) + decay (Green coproduct decomposition into lower states with conservation).

The quasi-eigentone framework gives the V3 movies a structural narrative: - A meson forms via Hall-algebra fusion (multiplication) of constituent quarks. - It's a QUASI-EIGENTONE (excited bulk state). - It decays via Green coproduct decomposition into lower K-types (other hadrons + leptons) with grading-conserved decay channels. - The “spectral score” visualizes each K-type’s Casimir; decays are arrows from higher to lower spectral levels.

7. Open items (L4 v0.2 and beyond)

This v0.1 framework places the STRUCTURAL distinction. Open multi-week items: - Quantitative decay rates — require the kernel matrix elements (Bergman integrals on the radial tower); L4 v0.2. - Branching ratios — the relative weights of different coproduct decay channels (e.g., why $W \rightarrow e\nu$ vs $\mu\nu$ have specific ratios). - CP violation in decays — CKM phase structure; CP-asymmetric coproduct (engine v0.4 baryogenesis link). - Hadron lifetime predictions — the full hadron spectrum’s coproduct decomposition structure (Grace’s catalog).

8. Honest scope + tier

RIGOROUS (engine theory): Green coproduct on Hall algebra; grading conservation; β -decay E3 as worked example.

FRAMEWORK (this v0.1): true/quasi/confined eigentone classification of SM particles based on K-type sector bottom-vs-excited structure. The structural distinction is rigorous; the per-particle classification rides on the keystone bet (K-types = particles, the dictionary’s load-bearing identification).

OPEN (multi-week): quantitative decay rates (kernel matrix elements), branching ratios, CP-violation in decays.

Cal #27 / honesty: this v0.1 places the structural DISTINCTION between stable and unstable particles — it is NOT a full derivation of decay rates. The classification is a clean structural reading of the dictionary’s K-type placements (which K-types are sector-bottoms = stable, which are excited = quasi); the per-particle quantitative dynamics are L4 v0.2 work. β -decay is the existing example; the framework generalizes the same coproduct structure to all SM decays.

Routed: → Casey/team: this is the structural framework that unifies SM stable-vs-unstable distinction under one engine-level mechanism (Green coproduct + grading conservation). Generalizes β -decay (E3) to the full SM. → Grace: candidate Periodic Table column — “stability class” (TRUE EIGENTONE / QUASI / EIGENTONE-IN-VACUUM) for every particle cell. → Elie: when bulk K-type radial tower lands (post-affine-pin), the kernel matrix elements give the quantitative decay rates. → Movie program: V0 spectral score + V3 formation/decay movies have this framework as their narrative spine.

— Lyra, Quasi-Eigentone Framework v0.1 (#397). The structural distinction: TRUE EIGENTONE = ground state in K-type sector (no allowed decay channels); QUASI-EIGENTONE = excited state with grading-conserved Green coproduct decay; EIGENTONE-IN-VACUUM = massless but confined (gluon). SM stable particles (e , ν , γ , p , gluon) = sector bottoms. SM unstable particles (μ , τ , n , W , Z , H , hadrons) = excited states with engine-level coproduct decay channels. β -decay (E3) is the first example; framework generalizes to all SM decays. Quantitative rates require L4 v0.2 kernel matrix elements. Movie program V0/V3 has its narrative spine.